

Week 7: Custom Dataset and Final Training Run

Theme:	Now make it yours.
Time Commitment:	10–12 hours (heavy compute week — start training early!)
Suggested Split:	2h reading, 4h data prep, 4–6h training (much of it unattended)

Learning Objectives

By the end of this week, you should be able to:

- Curate, clean, and preprocess a custom image dataset
- Make practical training decisions under realistic compute constraints
- Apply all techniques from previous weeks to a new domain
- Use experiment tracking tools to monitor and iterate on training runs

Primary Resources (Must-Do)

[BLOG/DOCS] Weights & Biases Quickstart

Type: Documentation | **Time:** 30 min | **Difficulty:** Beginner

URL: <https://docs.wandb.ai/quickstart>

Why this resource: Free experiment tracking that’s standard in industry. Logs your training curves, sample images, and hyperparameters automatically. After this week you’ll never go back to print statements.

Focus on: The PyTorch integration section. Add `wandb.log()` calls to your existing training loop.

[BLOG/DOCS] Hugging Face Datasets Library Tutorial

Type: Documentation + Code | **Time:** 45 min | **Difficulty:** Beginner-Intermediate

URL: <https://huggingface.co/docs/datasets/quickstart>

Why this resource: Easiest way to access thousands of pre-curated datasets and to upload your own.

Focus on: Loading image datasets, applying transforms, and creating PyTorch DataLoaders.

[BLOG/DOCS] Hugging Face: “Train a Diffusion Model” Tutorial

Type: Tutorial + Code | **Time:** 60 min | **Difficulty:** Intermediate

URL: https://huggingface.co/docs/diffusers/tutorials/basic_training

Why this resource: Walks through training on the Smithsonian Butterflies dataset — exactly the kind of small custom dataset you’ll be working with. Steal their data preprocessing patterns.

Focus on: The data loading and augmentation sections. The training loop is similar to yours.

[PAPER] EDM Paper: “Elucidating the Design Space of Diffusion-Based Generative Models” (Karras et al., 2022)

Type: Research Paper (selected sections) | **Time:** 30 min skim | **Difficulty:** Advanced
URL: <https://arxiv.org/pdf/2206.00364>

Why this resource: The most actionable training-tips paper in diffusion. The exact recommendations (preconditioning, augmentation, noise distribution) are worth knowing.

Focus on: Tables with training hyperparameters. Skip the theoretical reformulation unless interested.

Suggested Datasets

Pick one based on your interest:

- **Pokemon sprites** (~800 images, 64×64) — small enough to overfit creatively
- **Anime faces** (Danbooru subset, downsampled to 64×64)
- **CelebA faces** (downsampled to 64×64)
- **Quick, Draw!** sketches — pick one category, 28×28
- **LLD-icon** (logo dataset, 32×32)
- **Indian art / textile patterns** (scrape from open sources — culturally interesting and underrepresented)
- **Album covers, movie posters, book covers** (your interest — minimum 500 images, mentor approval required)

This Week’s Deliverable

Train your full diffusion model on a custom dataset. Code must include:

- A clean data preprocessing pipeline (resize, normalize to $[-1, 1]$, augment)
- A training script with W&B logging (loss, sample images, learning rate, FID if possible)
- At least 100 epochs of training (or until you see clear convergence)
- A **samples/** folder with generations at multiple training milestones (every 10–20 epochs)
- Final model checkpoint pushed to Hugging Face Hub (you’ll deploy from this in Week 8)
- A `TRAINING_LOG.md` documenting decisions made and lessons learned

Strategic tip: Start training Monday or Tuesday. Use the rest of the week to iterate, fix issues, and curate samples. Don’t leave training for the weekend.

Key Concepts Checklist

You should be able to confidently answer all of the following:

- How did you decide on image resolution? What trade-offs did you consider?
- What augmentations did you use? Why those specifically?
- How did you handle Colab disconnects during long training runs?
- What signs of mode collapse did you watch for?
- How did you decide when to stop training?
- What would you change if you had 10x the compute budget?

Stretch Resources (Optional)

- **Implement mixed-precision training** (`torch.cuda.amp`) for 1.5x speedup on T4 GPUs
- **Try a larger resolution** (128×128) and report on training time and quality
- **Compare to fine-tuned Stable Diffusion** on the same dataset using the `diffusers` library
- **Karras et al. EDM PyTorch reference:** <https://github.com/NVlabs/edm>

Debugging & Reference

- **Colab disconnect protection:** Save checkpoints every 10 epochs to Google Drive
- **Mode collapse symptoms:** All generated samples look identical or fall into 2–3 clusters — usually caused by training data issues, not model issues
- **Loss is decreasing but samples look bad:** Diffusion loss doesn't always correlate with visual quality. Check samples directly.
- **Out of memory errors:** Reduce batch size before reducing image resolution
- **Free Colab session ending:** Use Kaggle as backup (P100 GPU, 30 hours/week)